

APPLIED PHYSICS

High grain boundary recombination velocity in polycrystalline metal halide perovskites

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Understanding carrier recombination processes in metal halide perovskites is fundamentally important to further improving the efficiency of perovskite solar cells, yet the accurate recombination velocity at grain boundaries (GBs) has not been determined. Here, we report the determination of carrier recombination velocities at GBs (S_{GB}) of polycrystalline perovskites by mapping the transient photoluminescence pattern change induced by the nonradiative recombination of carriers at GBs. Charge recombination at GBs is revealed to be even stronger than at surfaces of unpassivated films, with average S_{GB} reaching 2200 to 3300 cm/s. Regular surface treatments do not passivate GBs because of the absence of contact at GBs. We find a surface treatment using tributyl(methyl)phosphonium dimethyl phosphate that can penetrate into GBs by partially dissolving GBs and converting it into one-dimensional perovskites. It reduces the average S_{GB} by four times, with the lowest S_{GB} of 410 cm/s, which is comparable to surface recombination velocities after passivation.

INTRODUCTION

Understanding carrier recombination behaviors in metal halide perovskite materials and devices is important for improving the performance of perovskite optoelectronic devices (1). Metal halide perovskites have achieved great progresses in photovoltaic application with rapid increase of the power conversion efficiency of perovskite solar cells from 3.8 to 25.5% (2–4) and other applications of radiation detectors (5, 6), light detectors (7, 8), and light-emitting diodes (9) with performance comparable to or better than existing technologies. While the bulk and surface recombination in perovskites have been widely studied, carrier recombination at the hidden grain boundaries (GBs) of polycrystalline perovskite films that are adopted in most of the state-of-the-art high-performance devices has not been well understood, leaving behind the ongoing debate of whether GBs are benign or detrimental to device performances (10).

Different studies have come to inconsistent conclusions on the role of GBs in determining the performance of perovskite solar cells. GBs were speculated to be benign without introducing mid-gap deep trap states (11–13), while they were also deemed to be nonradiative recombination centers, thus being detrimental to solar cells (10, 14, 15), or they were involved in a more complicated relationship with the solar cell performance, which was also related to the grain size and thickness of perovskite films (16–21). Conclusions of these studies, however, were mostly drawn empirically by investigating the relationship between solar cell performance and grain sizes, which could mistakenly confuse the effect of grain size changing-induced crystalline quality and point defects on device performance and not give precise descriptions about the carrier recombination at GBs. To date, there is still no direct or microscopic experimental method to quantitatively characterize the GB charge carrier recombination velocity (S_{GB}) in polycrystalline perovskites.

On the other hand, passivations of perovskite polycrystalline thin films on multilevels are shown globally to be indispensable to

achieve high-performance solar cells (22, 23). While the effect of surface/interface passivation for perovskite thin films has been well demonstrated, the often-speculated GB passivation by different kinds of passivators has not been rigorously verified because of the lack of a direct characterization technique. This raises an undetermined question of whether surface treatments of polycrystalline perovskites can also passivate defects at GBs.

In this work, we showed a method that can directly determine the S_{GB} of individual grains in polycrystalline perovskites. The influence of GB recombination on carrier diffusion in perovskite grains is visualized by the transient photoluminescence (PL) mapping (TPLM) pattern evolution induced by the charge recombination at GBs. A model is established to quantitatively describe the relationship between S_{GB} and carrier diffusion profiles. We evaluated the GB passivation effect for perovskite thin films with different surface treatment approaches, providing important guidelines for the further improvement of perovskite solar cell performance with desired passivation strategies.

RESULTS

Determination of GB recombination velocity

Figure 1A and fig. S1 show the schematic diagram of using confocal TPLM and carrier diffusion imaging system to determine S_{GB} . The S_{GB} is derived by measuring the carrier diffusion within a perovskite grain with the confocal system, where the evolution of PL intensity (I_{PL}), which is represented by carrier density n , can be described by a two-dimensional (2D) partial differential equation (24, 25)

$$\frac{\partial n}{\partial t} = D \nabla^2 n - An - Bn^2 - Cn^3 \quad (1)$$

where $n = n(x, y, t)$ is the spatial distribution of carrier density at time t ; A , B , and C are monomolecular, bimolecular, and Auger charge recombination rates, respectively; and D is the diffusion coefficient (detailed in Supplementary Text). Surface recombination may reduce the total recombination lifetime but does not affect the 2D carrier diffusion behaviors. In addition, it has been determined that reabsorption should have negligible influence on the lateral diffusion of carriers under pulsed excitation conditions for a perovskite

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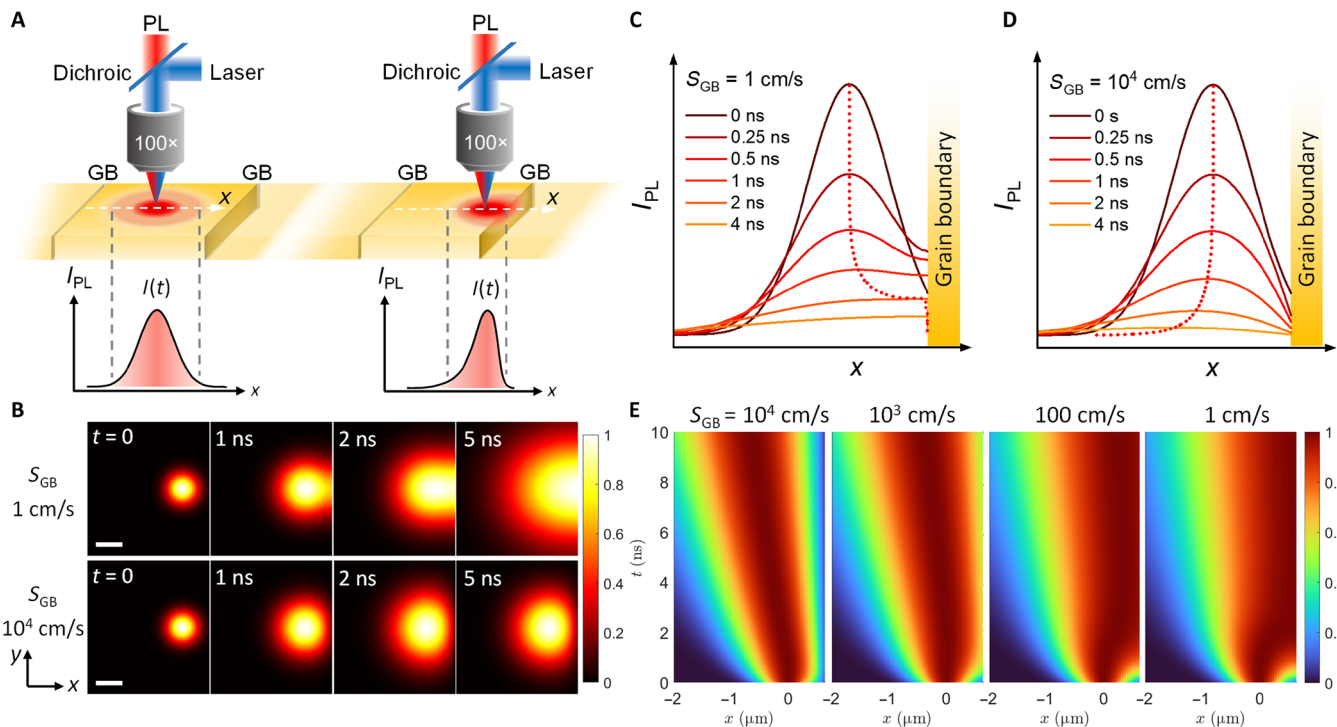


Fig. 1. Determination of GB charge recombination velocities. (A) Schematic diagram of using PL mapping and carrier diffusion imaging to measure GB recombination velocities. Excitation laser is focused onto the grain center (left) or near the GB (right) of perovskite films via a 100 $\times$ oil immersion objective. The carrier diffusion from the laser excitation spot is represented by the red ellipses. The linear PL intensity (I_{PL}) profiles along one direction (x), which is indicated by white dashed lines of the carrier diffusion image plane at time t , are representatively shown on the bottom. The gray dashed lines denote the edges of the PL profile curves. (B) Simulated 2D (x - y) carrier diffusion images of perovskite crystals with small (top) and large (bottom) S_{GB} at time frames of 0, 1, 2, and 5 ns. The I_{PL} is normalized to its peak value at each frame. Simulated linear PL profiles along one direction (x) of the perovskite crystals with (C) $S_{GB} = 1$ cm/s and (D) 10^4 cm/s at different times after laser excitation. The red dashed lines denote the shifts of the peak positions of the profile curves. The yellow blocks denote the locations of GBs. (E) Simulated linear carrier diffusion profiles along the normal direction of crystal edges of MAPbI₃ single crystals with different S_{GB} . The crystal edge is at the right edge of each figure.

material with PL quantum yield less than unity (24). This was further verified by measuring the PL spectra at both the grain center and GBs of perovskite films with laser excitation fixed at the grain center (fig. S2). The PL spectra measured at the grain center and GBs are basically identical without peak shifting or rising of PL emission at the low-energy side, and both exhibit rather symmetric spectral distributions with respect to photon energies, validating the negligible influence of reabsorption effect during the carrier diffusion measurement. Two different boundary conditions are considered depending on the distance (L_{GB}) of the laser excitation spot to the GB. When L_{GB} is larger than the diffusion length of carriers L_D , almost all the PL signal is collected before the photoexcited carriers reach the GBs, and an isotropic carrier diffusion behavior is guaranteed. The boundary condition will be

$$\vec{n} \cdot \nabla n(x, y, t) = 0 \quad (2)$$

where $\vec{n}$ is the outward unit normal vector. Otherwise, the boundary condition involving S_{GB} of

$$\vec{n} \cdot \nabla n(x, y, t) + \frac{S_{GB}}{D} n(x, y, t) = 0 \quad (3)$$

is applied. In this case, GB recombination should cause anisotropic carrier diffusions along the normal direction of the GB, leading to asymmetric PL profile distributions (Fig. 1A). We should mention

that these analyses are established on the basis of experimental observations that carrier diffusions are blocked by GBs in most of the grains (fig. S3), which is a common observation reported in literature for most perovskite thin films (26–29). However, we also noticed that for about 15% of the grains that we have measured, carriers could pass through the GBs, similar to what has been reported in a few works (30, 31). It is still unclear why carriers can pass through these GBs, while this work mainly focuses on the GBs that block the intergrain carrier diffusions.

To illustrate how S_{GB} affects carrier diffusion behaviors, we simulated the dynamic process of carrier diffusion near the GBs with small and large S_{GB} based on Eqs. 1 and 3 (detailed in Supplementary Text). The initial carrier concentration distribution was implemented with a 2D Gaussian function, simulating the light distribution of a laser spot (fig. S1B). Movies S1 and S2 show the dynamic diffusion process for $S_{GB} = 1$ and 10^4 cm/s, respectively, normalized to the peak value for each frame. Figure 1B shows selected frames from the videos. The light spots expand asymmetrically along different in-plane directions accompanied by a shifting of emission peak positions, indicating anisotropic carrier diffusions near the GB due to the influence of charge recombination at GBs. Figure 1 (C and D) plots the detailed PL intensity profiles at different time delays after laser excitation along the x direction as labeled in the figure for $S_{GB} = 1$ and 10^4 cm/s. When S_{GB} is small (1 cm/s), the PL intensity profile asymmetrically broadens toward the GB with its peak

shifting toward the GB (Fig. 1C and fig. S4A). This can be explained by the reduced carrier density at the GB side due to the moderate charge recombination, generating a descending carrier density gradient toward the GB, which drives the diffusion of excess carriers to the GB. A small S_{GB} implies a weak carrier recombination and long-living carriers, enabling a carrier density accumulation at the GB (Fig. 1C and fig. S4A). In contrast, when S_{GB} is sufficiently large (10^4 cm/s), the carrier density at the GB is almost zero at any time (Fig. 1D and fig. S4C). In this case, the peak of the PL intensity profile shifts away from the GB because of the fast quenching of charge carriers at GBs that drags down the carrier concentration close to the GBs. When S_{GB} is in the intermediate range, e.g., 10 to 1000 cm/s, there is a transition for the peak from moving toward the GB to back-shifting inward. This resulted from the competition between the accumulation and recombination of carriers on the GB side. Upon laser excitation, carriers diffuse and accumulate at the GB side if the charge recombination at the GB is weak, resulting in the PL center shifting toward the GB. If charge recombination at the GB is fast, then the charges close to GBs quickly deplete, resulting in a faster reduction of the carrier density on the GB side and shifting of the PL center away from GBs (fig. S4B). This simulation result shows that the charge recombination at GBs can cause the PL intensity pattern change, which provides a quantifiable method to determine S_{GB} by fitting the measured PL intensity profiles near the GB (Fig. 1E).

To verify it experimentally, we first determine the parameters D , A , B , and C by placing the light excitation spot at the center of a grain and then moving the light excitation spot close to the GBs ($L_{GB} < 1 \mu\text{m}$) to measure PL intensity profile variation over time using transient PL mapping measurement. In this measurement, the spatial distribution of PL at the sample surface is projected onto an image plane through a telescopic relay lens system and scanned by an optical fiber mounted on a precise x - y piezo motor stage, which guides light into a single-photon photodetector (see Materials and Methods). By measuring the carrier density propagation from the grain center, we derived D from the relationship $\sigma^2(t) = 2Dt + \sigma_0^2$, where $\sigma^2(t)$ is the mean square distribution of the Gaussian distribution-fitted PL profile along one propagation direction at the time delay t after the laser excitation and σ_0^2 is the initial mean square distribution at $t = 0$ (32–34). By fitting the measured confocal time-resolved PL (TRPL) and PL profiles with Eq. 1 and boundary condition (Eq. 2), we obtain recombination rate constants A , B , and C . The S_{GB} is then calculated by fitting the measured carrier diffusion profiles along the normal direction of the GB with simulations based on Eq. 1 and boundary condition (Eq. 3).

We first used the edges of a single crystal to represent GBs and measured the surface recombination velocities (S_{surf}) of freshly cleaved methylammonium lead iodide (MAPbI₃) and methylammonium lead bromide (MAPbBr₃) single crystals (as illustrated in Fig. 2A) because there are other reported methods to measure S_{surf} that can be used to verify this method. We also compared the S_{surf} of MAPbBr₃ single crystals with and without ozone treatment (MAPbBr₃-O₃), which was reported obviously to modify the S_{surf} (35). As shown in figs. S5 to S7, the diffusivities for MAPbI₃ and MAPbBr₃ crystals measured by the diffusion method are 0.67 and 0.49 cm²/s, respectively, which are close to previously reported values measured by TPLM or time-of-flight methods (36, 37). By fitting the TRPL and PL profile results, we obtained the values of A , B , and C for MAPbI₃ and MAPbBr₃ single crystals, as summarized in fig. S8

and table S1. Next, we moved the laser excitation spots to near the crystal edge with $L_{GB} = \sim 0.6 \mu\text{m}$ and performed the same carrier diffusion measurement. Figure 2 (B and C) shows the measured carrier diffusion profile and PL profiles along the normal direction of the crystal edge for the MAPbI₃ single crystal, respectively. Movie S3 shows the dynamic PL spreading process by carrier diffusion. For the MAPbI₃ single crystal, the PL profile peak gradually shifted to the crystal edge after laser excitation, indicating an accumulation of carriers at the crystal edge and thus a small S_{surf} . For MAPbBr₃ single crystals with fresh surface, the peak of the PL profile only slightly merged to the crystal edge after laser excitation and then moved away from the surface (Fig. 2, E and F). The measured trace of the peak shifting along the normal direction of the crystal surface can be well fitted with the simulated curve with $S_{GB} = 102 \pm 45$ cm/s for MAPbI₃ crystals and 400 ± 190 cm/s for MAPbBr₃ crystals with fresh surfaces (Fig. 2, D and G), showing that the bromide crystals have much stronger surface charge recombination. Statistic results of S_{GB} and simulations with different D and L_{GB} are shown in figs. S9 to S12. After ozone treatment, the average S_{surf} of MAPbBr₃ is reduced to 150 ± 40 cm/s (Fig. 2, H to J, and fig. S13), which is consistent with the previous discovery that O₃ could effectively passivate the MAPbBr₃ single crystal surface (35). We then measured S_{surf} of the MAPbI₃, MAPbBr₃, and MAPbBr₃-O₃ single crystals with the reported surface TRPL measurement that conducted directly on the surface of the crystals (see Supplementary Text and fig. S14). The S_{surf} is derived from surface recombination lifetime τ_s (38, 39)

$$\frac{1}{\tau_s} = \frac{2\alpha^2 D}{1 + \sqrt{1 + 8\alpha^2 D^2 / S_{\text{surf}}^2}} \quad (4)$$

where α is the absorption coefficient at the excitation wavelength. The derived S_{surf} values are 60, 650, and 210 cm/s for MAPbI₃, MAPbBr₃, and MAPbBr₃-O₃ single crystals, respectively, which are close to those obtained from our diffusion method. This validates the effectiveness of our method in determining the S_{GB} by charge recombination-induced transient PL pattern change.

GB recombination velocity of polycrystalline films

We then investigated the S_{GB} of polycrystalline perovskite thin films. We chose MAPbI₃ and Cs_{0.08}FA_{0.92}PbI₃ (FA is formamidinium) thin films for this study, which are two very typical compositions that showed high solar cell device performance (40). We mainly chose grains with lateral sizes $> 3 \mu\text{m}$ so that the carrier diffusion near the studied GBs was not affected by adjacent GBs within the time range of measurement (e.g., 20 ns). The interplay of multiple GBs could make a difference to the carrier diffusion behavior (24), although it is not within the scope of the current study. Figure 3A shows the PL mapping image of a pristine MAPbI₃ thin film. We first measured the D of individual grains by focusing the laser spot onto the grain center (inset of Fig. 3A). Figure 3 (B and C) shows the measured and simulated carrier density profiles and PL profiles for a pristine MAPbI₃ grain, respectively. To rule out the possible influence from the un-recombined background carrier density accumulated in the space-limited grains on the Gaussian fitting of the PL profiles, we checked the accuracy of the PL profile fitting obtained with different laser repetition rates, as shown in fig. S15. A laser repetition rate of up to 20 MHz was chosen to ensure a negligible influence from the background carrier density on the calculation of D while retaining a high carrier population rate to minimize the measurement duration. In addition, we checked the light stability of the perovskite films, of

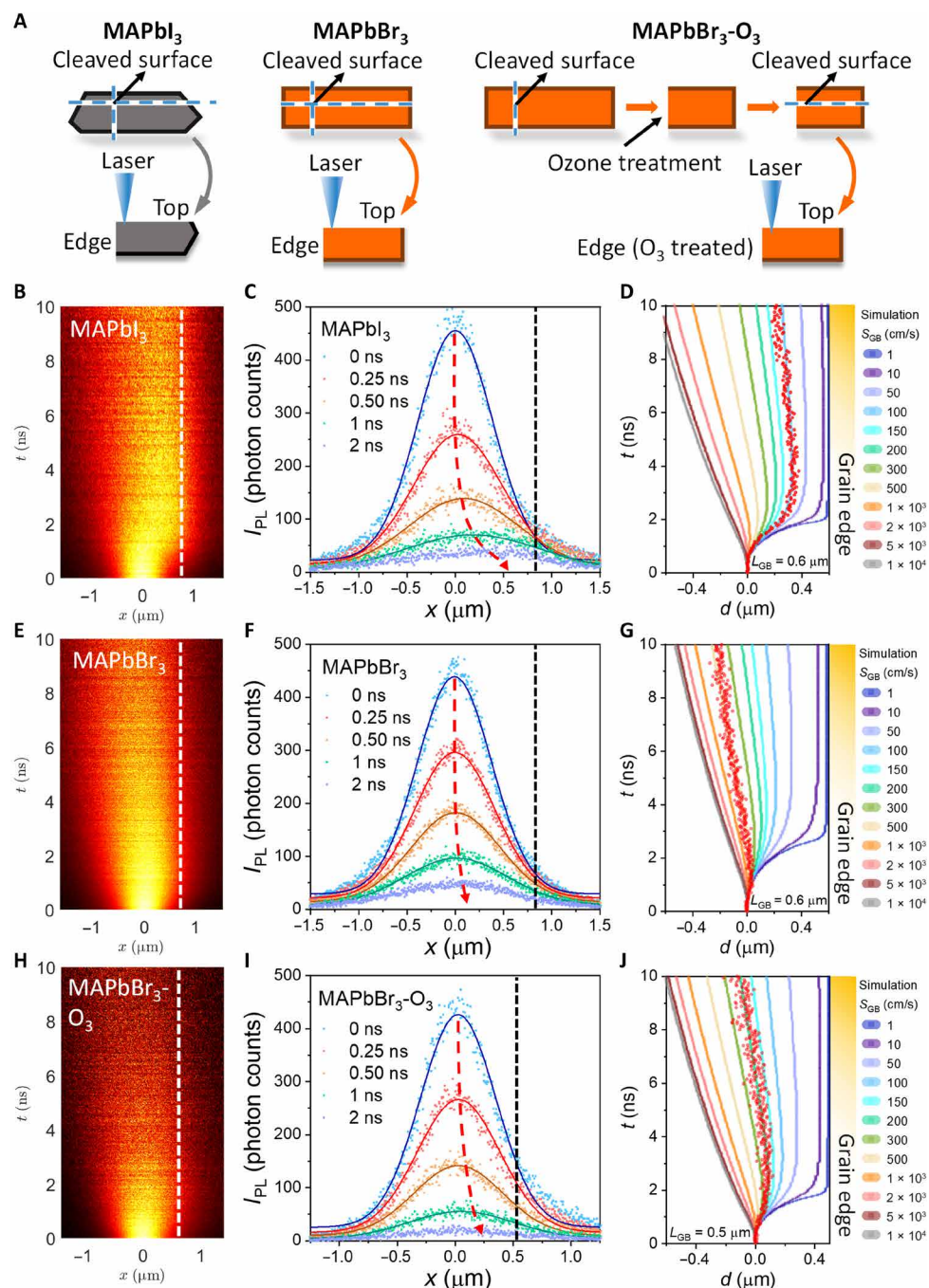


Fig. 2. Surface charge recombination velocities of perovskite single crystals. (A) Schematic illustrations of sample preparation processes for MAPbI₃, MAPbBr₃, and ozone-passivated MAPbBr₃ (MAPbBr₃-O₃) single crystals for the measurement of S_{surf} . Note that only the crystal edge of MAPbBr₃-O₃ was treated by ozone. Measured carrier diffusion profiles near the crystal edges of the (B) MAPbI₃, (E) MAPbBr₃, and (H) MAPbBr₃-O₃ single crystals. The white dashed lines denote the locations of crystal edges. Measured PL intensity profiles near the crystal edges of the (C) MAPbI₃, (F) MAPbBr₃, and (I) MAPbBr₃-O₃ single crystals. The black dashed lines denote the locations of crystal edges. The red dashed lines indicate the peak shifts of the PL profile curves with time. Traces of the peak shifting with time extracted from (B), (E), and (H) (red dots) overlapped with those obtained from simulation results with different S_{GB} (colored lines) for (D) MAPbI₃, (G) MAPbBr₃, and (J) MAPbBr₃-O₃ single crystals.

which the PL intensity change was less than $\pm 5\%$ within the measurement time of 20 to 30 min (fig. S16).

The average D values derived from statistic measurements on multiple pristine MAPbI₃ and Cs_{0.08}FA_{0.92}PbI₃ grains are ~ 1.0 and $1.1 \text{ cm}^2/\text{s}$, respectively (figs. S17, A and B; S18, A and B; and S21A

and tables S2 and S3). Next, we performed different surface treatments to the MAPbI₃ and Cs_{0.08}FA_{0.92}PbI₃ thin films, including (C8-NH₃)₂SO₄ (sulfate) and tributyl(methyl)phosphonium dimethyl phosphate (TPDMP). Passivating perovskite surface by forming a lead sulfate layer with sulfate treatment (41, 42) or forming phosphonium

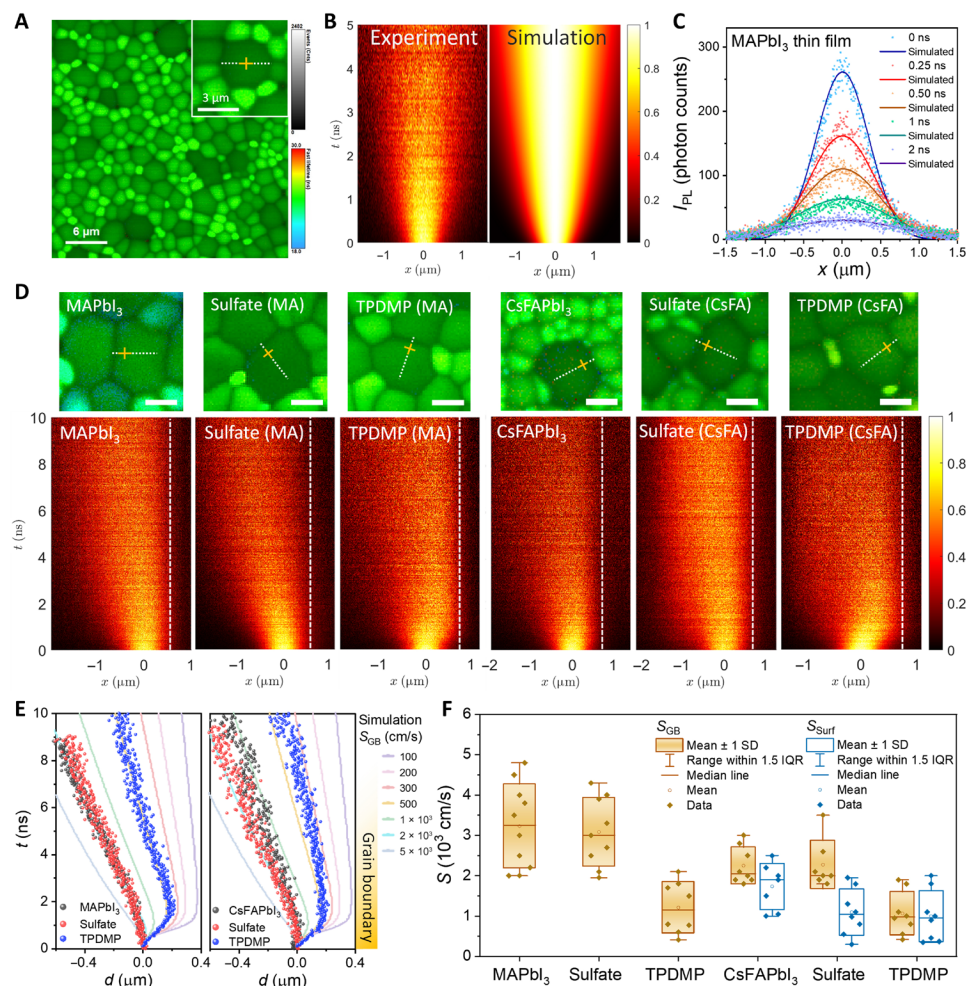


Fig. 3. GB charge recombination velocities of perovskite thin films with different passivation. (A) TRPL mapping image of a MAPbI₃ polycrystalline thin film. The inset shows the zoomed-in PL mapping image of a MAPbI₃ grain. The yellow cross denotes the laser excitation spot, and the white dashed line denotes the direction for the carrier diffusion and PL profiles. (B) Measured (left) and simulated (right) linear carrier diffusion profiles of a MAPbI₃ grain in the thin film when the laser excites the grain center. (C) Measured (dots) and simulated (solid lines) linear PL profiles at different times after excitation of the MAPbI₃ grain when the laser excites the grain center. (D) PL mapping images of pristine, sulfate-treated, and TPDMP-treated MAPbI₃ and pristine, sulfate-treated, and TPDMP-treated Cs_{0.08}FA_{0.92}PbI₃ grains (top) chosen for carrier diffusion measurement and the corresponding carrier diffusion profiles along the normal direction of GBs of these grains (bottom). The yellow crosses in the PL mapping images denote the laser excitation spots that are around 0.6 μm away from the GBs. The white dashed lines in the diffusion profile images denote the locations of GBs. Scale bars, 2 μm. (E) Overlapping of the measured traces of peak shifting of carrier diffusion profiles extracted from (D) (dots) with those obtained from simulation results for MAPbI₃ (left) and Cs_{0.08}FA_{0.92}PbI₃ (right) grains with different S_{GB} (solid lines). The yellow block denotes the location of the GB. (F) Statistic results of S_{GB} of grains in pristine, sulfate-treated, and TPDMP-treated MAPbI₃ and Cs_{0.08}FA_{0.92}PbI₃ thin films, together with the S_{surf} of pristine, sulfate-treated, and TPDMP-treated Cs_{0.08}FA_{0.92}PbI₃ thin films.

lead halide compounds using phosphonium salts (43, 44) has been previously demonstrated to reduce surface defects for perovskites. The phosphate-base anion in TPDMP is chosen to further passivate undercoordinated lead ions by forming ionic bonding. After sulfate and TPDMP treatments, both the MAPbI₃ and Cs_{0.08}FA_{0.92}PbI₃ thin films exhibited substantially enhanced surface PL intensities (fig. S22, A and B). However, it is unknown whether these passivation agents can reach and passivate GBs as well. To uncover this puzzle, we then characterize the S_{GB} of MAPbI₃ and Cs_{0.08}FA_{0.92}PbI₃ thin films before and after sulfate and TPDMP treatments.

First, we noticed that these surface treatments do not affect the D value for MAPbI₃ and Cs_{0.08}FA_{0.92}PbI₃ thin films (figs. S17, C to F; S18, C to F; and S21A and tables S2 and S3), which is reasonable

because surface treatments should not change the bulk properties of the perovskite grains. By fitting the recombination rate constants A, B, and C from the confocal TRPL and PL profile results (figs. S19 and S20), it is seen that the bimolecular charge recombination rate remained unchanged before and after surface treatments of both MAPbI₃ and Cs_{0.08}FA_{0.92}PbI₃ because this is an intrinsic material property for perovskites. In contrast, the trapping-related charge recombination is smaller for surface-treated films, indicating that these surface treatments did passivate the surface of the perovskite thin films (fig. S21, B and C, and tables S2 and S3).

Then, we measured the carrier diffusions near the GBs for both MAPbI₃ and Cs_{0.08}FA_{0.92}PbI₃ thin films before and after surface treatments. Typical carrier diffusion profiles along the normal directions

of the GBs and their corresponding PL mapping images are shown in Fig. 3D. Statistic results of the carrier diffusion profiles for multiple grains are shown in figs. S23 to S28. We can see that for pristine and sulfate-treated MAPbI₃ and Cs_{0.08}FA_{0.92}PbI₃ thin films, the peaks of the diffusion profiles monotonously shift to the center of the grain after laser excitation, indicating a strong GB recombination in these films, while for TPDMP-treated films, the peak first shifted toward the GBs and then moved to the grain center. By fitting the traces of the peak shifting for these films with simulation results, we obtain the S_{GB} for each film, as shown in Fig. 3 (E and F). For pristine MAPbI₃ thin films, the S_{GB} varies between 2000 and 5000 cm/s for different grains with an average S_{GB} of 3300 ± 1040 cm/s. The large deviation of the S_{GB} values for different grains indicates a large grain-to-grain heterogeneity in polycrystalline perovskite thin films in terms of GB recombinations (45). Sulfate treatment does not significantly reduce the GB recombination, as it renders a similar average S_{GB} of 3100 ± 950 cm/s. In contrast, TPDMP surface treatment reduces the S_{GB} to 1200 ± 640 cm/s, with the lowest S_{GB} of about 410 cm/s for about 25% of grains measured. Similar results were obtained for Cs_{0.08}FA_{0.92}PbI₃ thin films. The pristine and sulfated-treated films have an S_{GB} of 2200 ± 460 cm/s, while TPDMP-treated films show a much smaller S_{GB} of 900 ± 490 cm/s. These results demonstrate that TPDMP surface treatment can also passivate GBs for polycrystalline perovskite thin films, which is very different from sulfate treatment.

GB passivation mechanism

To understand why TPDMP can passivate the GBs of perovskite polycrystalline films, we first performed PL and atomic force microscopy-infrared (AFM-IR) measurements to check whether these passivation agents could really reach the GBs. As shown in fig. S22 (C and D), the PL intensity from the bottom side was basically unchanged in sulfate-treated films, while it was substantially enhanced in TPDMP-treated films, indicating that TPDMP did infiltrate into the perovskite films after surface treatment. For the AFM-IR measurement, we chose IR signals at wave numbers of 965 cm^{-1} , which corresponds to C—H bending vibrations in TP ions, and 1713 cm^{-1} , which corresponds to C=N stretching in FA to distinguish the IR response from the passivation agents/reaction production and Cs_{0.08}FA_{0.92}PbI₃ perovskite (fig. S29) (46). We did not use MAPbI₃ for the IR measurement because of the overlapping of C—H vibration IR signals between MA and TP ions. As shown in Fig. 4 (A and B), for TPDMP-treated Cs_{0.08}FA_{0.92}PbI₃ films, both the GBs and grain surfaces show high IR signals at a wave number of 965 cm^{-1} , while the GBs have much lower IR signals at a wave number of 1713 cm^{-1} compared to the grain surfaces. In contrast, the pristine films show negligible IR signals around GBs at all the wave numbers (Fig. 4A). This indicates that TP ions not only cover grain surface but also penetrate into the GBs of polycrystalline perovskite films and accumulate there. Then, we further revealed the GB passivation mechanism by investigating the chemical reactions between TPDMP and perovskites. As shown in Supplementary Text and fig. S30, TPDMP can readily react with perovskite and dissolve it at room temperature. The top surface PL intensities of perovskite films decreased after coating of TPDMP without annealing and then enhanced after annealing. X-ray diffraction (XRD) results show that the reaction between TPDMP and perovskite at room temperature generates an intermediate phase first, which eventually turned into TPPbI₃ after annealing (fig. S30). The PL intensity change at the bottom

interface followed the same trend as the top surface, suggesting that the TPDMP penetrated through the whole perovskite films. This is further supported by the unchanged PL intensity for the sulfate surface treatment, which only change the top surface (fig. S22). These results reveal that the penetration of TP ions into perovskites along GBs occurred by the quick conversion of 3D perovskite surface and GBs into a liquid intermediate phase first, and then, the intermediate phase is converted to a 1D perovskite via the reaction during annealing at 100°C



which can passivate the GBs, as schematically illustrated in Fig. 4C.

GB recombination versus surface recombination

To compare the GB and surface passivation capabilities of TPDMP, we directly measured the S_{surf} of Cs_{0.08}FA_{0.92}PbI₃ films before and after surface treatment by the same carrier diffusion measurements. To do this measurement, we need to rotate the sample and measure the cross sections of these films with the geometry shown in Fig. 4D. Figure 4E shows the PL mapping images for the cross sections of pristine, sulfate-treated, and TPDMP-treated Cs_{0.08}FA_{0.92}PbI₃ thin films deposited on glass. We focused the laser spot near the top surface of the grains for each film and performed the carrier diffusion measurement. Figure 4F and fig. S32 show the traces of peak shifting along the normal direction of the grain surface for these films. Statistical S_{surf} values are summarized in Fig. 3F. It is noteworthy that the S_{surf} was smaller than the S_{GB} for pristine Cs_{0.08}FA_{0.92}PbI₃ films (1800 versus 2200 cm/s), indicating that GBs are even more detrimental to carrier recombination than the surface for Cs_{0.08}FA_{0.92}PbI₃ films. After sulfate treatment, the S_{surf} is reduced to 1000 ± 490 cm/s, validating the surface passivation effect by sulfate treatment. For the TPDMP-treated films, the average S_{surf} is similar to S_{GB} , as both are around 900 cm/s. This highlights that TPDMP has good passivation effects for both the grain surfaces and GBs.

After knowing the S_{surf} and S_{GB} for perovskite single crystals and polycrystalline thin films, we further estimated the surface and GB interfacial trap densities (N_t) by using $S = N_t \sigma v_{\text{th}}$ (where σ is the capture cross section for electrons or holes, which is estimated to be 10^{-15} cm^2 , and v_{th} is the thermal velocity, which is 10^7 cm/s) (47–49). As shown in Fig. 4G, the surface trap densities of MAPbI₃ and MAPbBr₃ single crystals are around 10^{10} cm^{-2} , while those of polycrystalline thin films are over 10^{11} cm^{-2} . For pristine polycrystalline thin films, the trap densities at GBs of about $3 \times 10^{11}\text{ cm}^{-2}$ are slightly higher than that at grain surfaces.

Fast quantification of GB recombination velocity

To make this method easy to use, we develop a fast quantification method to derive S_{GB} for perovskites by using one simplified parameter, d_{max}/L_{GB} , where d_{max} is the maximum displacement of the carrier profile peak shifting toward the GB, as schematically illustrated in Fig. 5A. The dependence of the normalized peak shifting d_{max}/L_{GB} on S_{GB} for our perovskites with D varying from 0.1 to $2\text{ cm}^2/\text{s}$ is plotted in Fig. 5B. This provides a quick estimation of S_{GB} (applies for S_{surf} as well) for most perovskites by simply measuring the relative displacement of the carrier profile peak shifting with respect to the total distance to GBs. To verify the applicability of this quantification method for perovskites, we checked the influences of

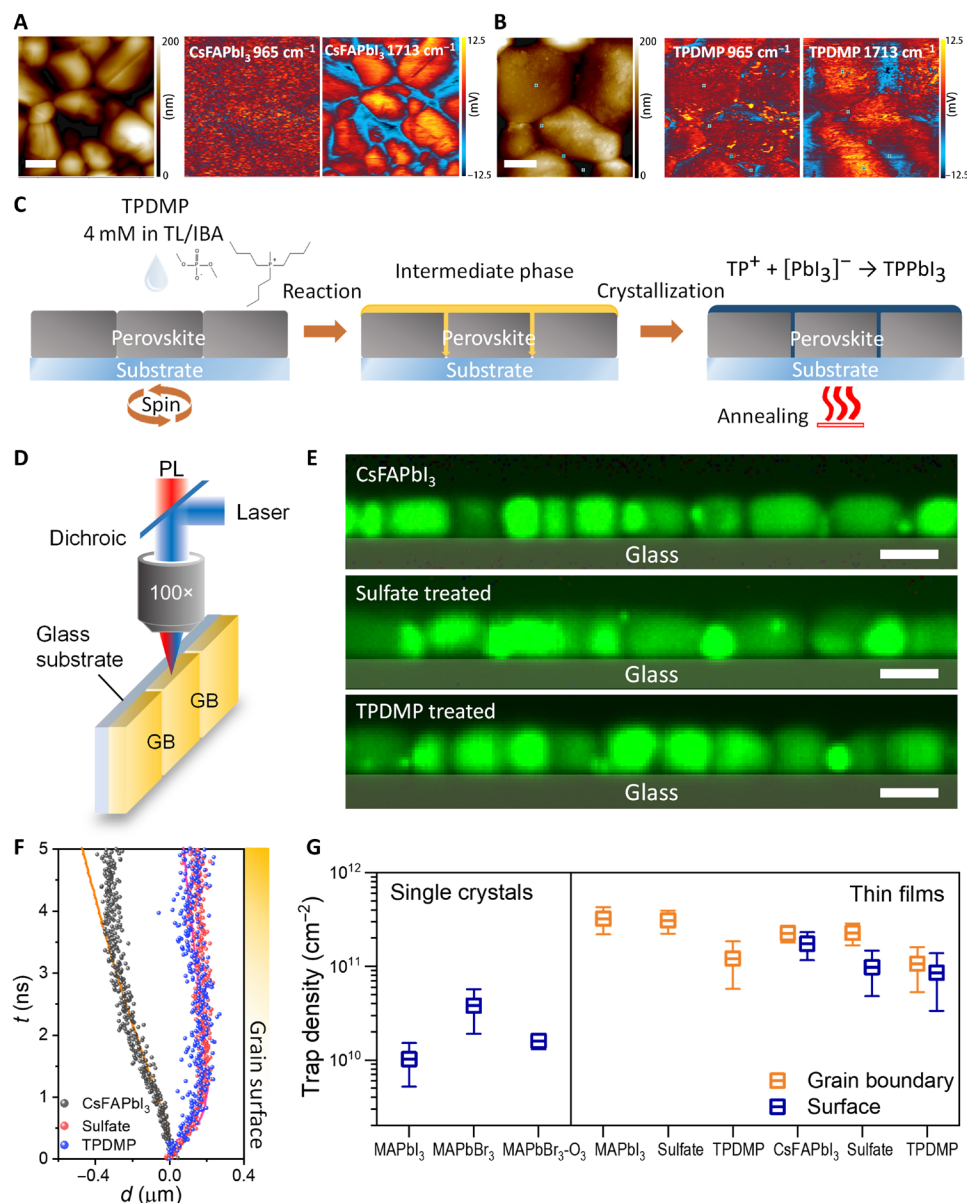


Fig. 4. GB passivation of perovskite thin films by TPDMP. AFM topography (left) and IR images of (A) pristine and (B) TPDMP-treated Cs_{0.08}FA_{0.92}PbI₃ polycrystalline thin films measured at wave numbers of 965 cm⁻¹ (middle) and 1713 cm⁻¹ (right). Scale bars, 2 μm. (C) Schematic illustration of the GB passivation process by TPDMP. (D) Schematic diagram of using PL mapping and carrier diffusion imaging to measure the cross section of polycrystalline perovskite thin films. (E) TRPL mapping images of cross sections of pristine, sulfate-treated, and TPDMP-treated Cs_{0.08}FA_{0.92}PbI₃ thin films. Scale bars, 2 μm. (F) Overlapping of the measured traces of peak shifting of carrier diffusion profiles with laser excitations near the grain surfaces of pristine, sulfate-treated, and TPDMP-treated Cs_{0.08}FA_{0.92}PbI₃ thin films with simulation results for S_{GB} = 2000 cm/s (orange line) and 300 cm/s (pink line). The yellow block denotes the location of the grain surface. (G) Estimated surface and GB trap densities of perovskite single crystals and polycrystalline thin films.

L_{GB} and recombination rate coefficients A and B on the quantitative relationship between d_{max}/L_{GB} and S_{GB} , as shown in figs. S33 and S34. It is seen that the variations of L_{GB} , A , and B from 0.4 to 0.8 μm, 0.2 to 1 ns⁻¹, and 10⁻⁴ to 2 × 10⁻³ cm²/s, respectively, which safely cover the ranges of all the measured values for perovskites in this work, have negligible influence on the dependence of d_{max}/L_{GB} on S_{GB} , thus validating the universal application of Fig. 5B for most perovskites.

DISCUSSION

In summary, we establish a characterization method that is able to determine the GB recombination velocity at microscopic scales in polycrystalline perovskite thin films and evaluate the passivation effect at the hidden GBs by different passivation agents. We demonstrate that regular surface passivation methods such as sulfate treatment can only passivate grain surface while not GBs, making GB detrimental to solar cell performance. A new surface passivation

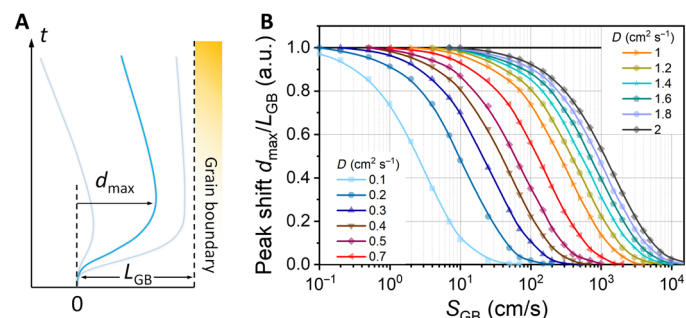


Fig. 5. Fast quantification of GB charge recombination velocities in perovskites.

(A) Schematics of the variation of the trace of carrier diffusion profile peak shifting with S_{GB} . The location 0 indicates the laser excitation spot. The bright and dim blue lines plot the traces of the peak shifting with different S_{GB} . d_{max} is the maximum displacement of the carrier profile peak shifting toward the GB. (B) Dependence of normalized peak shifting d_{max}/L_{GB} on S_{GB} for perovskites with D varying from 0.1 to 2 cm²/s. This quantitative relationship is valid for L_{GB} , A , and B ranging from 0.4 to 0.8 μ m, 0.2 to 1 ns⁻¹, and 10^{-4} to 2×10^{-3} cm²/s, respectively. a.u., arbitrary units.

method using TPDMP can passivate both grain surface and boundaries, pointing out a promising future direction for the further improvement of perovskite solar cells by passivating GBs via simple surface treatments.

MATERIALS AND METHODS

Materials

Unless stated otherwise, all the materials and solvent were purchased from Sigma-Aldrich. Lead iodide (PbI_2) with the purity of 99.995% was purchased from BeanTown Chemical. MAI and MABr were purchased from GreatCell Solar.

Synthesis of perovskite single crystals

The $MAPbI_3$ and $MAPbBr_3$ single crystals were synthesized by inverse temperature crystallization method. The precursor solution for the synthesis of $MAPbI_3$ single crystals was prepared by dissolving MAI and PbI_2 with the molar ratio of 1:1 in γ -butyrolactone at a concentration of 1.5 M. The solution was heated at 65°C with magnetic stirring until the solutes fully dissolved, followed by filtering using a polytetrafluoroethylene filter with a 0.2- μ m pore size and injected into a vial. The vial was then placed on a hot plate at the temperature of 75°C. The temperature was gradually increased to boost the growth of the single crystal. The $MAPbBr_3$ single crystals were synthesized in the same procedure with that for the $MAPbI_3$ single crystals. $MAPbBr_3$ precursor solution (1.2 M) was prepared by dissolving MABr and $PbBr_2$ with the molar ratio of 1:1 in dimethylformamide (DMF) solvent and then added into a vial on the hot plate for the crystallization of the single crystals. For the ozone treatment of the $MAPbBr_3$ single crystal, the crystal was first cleaved to expose fresh surfaces and then treated with ultraviolet (UV) ozone for 20 min.

Fabrication of polycrystalline perovskite thin films

For the fabrication of $MAPbI_3$ polycrystalline thin films, glass substrates were first prepared by cleaning with detergent, deionized water, acetone, and isopropanol in sequence and UV ozone treatment (20 min). $MAPbI_3$ precursor solution (1.4 M) was prepared by

dissolving MAI and PbI_2 with the molar ratio of 1:1 in DMF and dimethyl sulfoxide (DMSO; with the volume ratio of 9:1). A 1.5% volume ratio of methylammonium hypophosphite (MAH_2PO_2) was then added to the precursor solution to tune the grain size of $MAPbI_3$ thin films. The precursor solution was spin-coated onto the glass substrate at 2000 rpm for 2 s and then 4000 rpm for 20 s. During the spin coating, 130 μ l of toluene was added into the film at the eighth second. The as-formed film was then heated at 100°C for 10 min and at 120°C for 10 min. The $Cs_{0.08}FA_{0.92}PbI_3$ perovskite films were fabricated by blade coating method at room temperature under nitrogen knife blowing by using a precursor solution containing 1.0 M $FAPbI_3$ and 0.09 M $CsPbI_3$ dissolved in a 2-methoxyethanol:DMSO solvent mixture. A 1.5% volume ratio of MAH_2PO_2 was added to the precursor solution to tune the grain size of $Cs_{0.08}FA_{0.92}PbI_3$. The as-coated solid film was annealed at 150°C in air for 2 min. The oxy salt treatment of the polycrystalline perovskite thin films was carried out by spin coating 2 mM $(C_8-NH_3)_2SO_4$ solution in toluene/isopropyl alcohol mixture on the as-fabricated thin films at 5000 rpm for 30 s, followed by thermal annealing at 100°C for 10 min. For the TPDMP treatment, TPDMP solution was prepared by dissolving TPDMP in a toluene : isobutanol = 1:4 solution with a concentration of 4 mM and followed by dynamic spin coating onto the perovskite thin films at 5000 rpm for 30 s. After the spin coating, the film was annealed at 100°C for 5 min. For the fabrication of polycrystalline thin-film solar cells, a poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine] solution with the concentration of 2 mg/ml in toluene was spin-coated onto the indium tin oxide glass substrate at 4000 rpm for 30 s, followed by the fabrication of perovskite thin films. The whole device was completed by sequentially thermal evaporating C_{60} (30 nm), bathocuproine (6 nm), and Cu electrodes (80 nm) on the perovskite film.

PL mapping and carrier diffusion measurements

TPLM and carrier diffusion imaging were measured by a PicoQuant Micro Time 100 confocal microscopic system. An Olympus 100 $\times$ oil immersion objective with a numerical aperture of 1.45 was used for the measurement. Samples were covered by immersion oil (Olympus Immersion Oil Type-F) in a dark environment during the measurement. Picosecond 640- and 480-nm pulsed lasers with a repetition rate of 20 MHz were used to measure $MAPbI_3$ and $MAPbBr_3$ perovskites, respectively. The focused lasers generated near-Gaussian distributions of light intensities, with full width at half maximum of about 510 and 440 nm for the 640- and 480-nm laser sources, respectively. The average laser intensity was kept at around 36 nW at 20 MHz, corresponding to a photon energy of 1.8 fJ per pulse and an approximate photon flux of 6000 photons per pulse. The PL generated by samples was collected through the same objective and relayed onto an image plane through a telescopic relay lens system. The transmitted epifluorescence image was then collected by a single-photon detector (PicoQuant PMA Hybrid) through an optical fiber that was connected to a precise x - y piezo motor stage (PI U-751.24) with a linear step resolution of 100 nm. The minimum temporal resolution of the system was 25 ps, which was determined by the time-correlated single-photon counting modules (TimeHarp 260). Statistical results were collected from more than two batches of samples with more than 15 grains for each type of perovskite thin film. The one-photon surface TRPL and PL spectrum measurements were conducted with a PicoQuant Fluo Time 300 fluorescence lifetime and steady-state spectrometer. A 405-nm pulsed laser with

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